

CS 171 / EE 142: Intro to ML and DM

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UC Riverside

Slide Set 9: Regularization



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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 - ▶ For use only by enrolled students in the course

Regularization

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More generally:

$$L = \underbrace{\sum_{i=1}^m l(y_i, f(\vec{x}_i))}_{\text{loss on all training data}} + \underbrace{\lambda R(\vec{w})}_{\substack{\text{regularizer} \\ \uparrow \\ \text{regularization strength}}} .$$

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Logistic Regression with ℓ_2 Regularization

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$$\vec{w} \leftarrow \vec{w} + \eta \left(\sum_{i=1}^m (1 - \sigma(y_i \vec{x}_i^\top \vec{w})) y_i \vec{x}_i \right) - 2\eta \lambda \vec{w}$$

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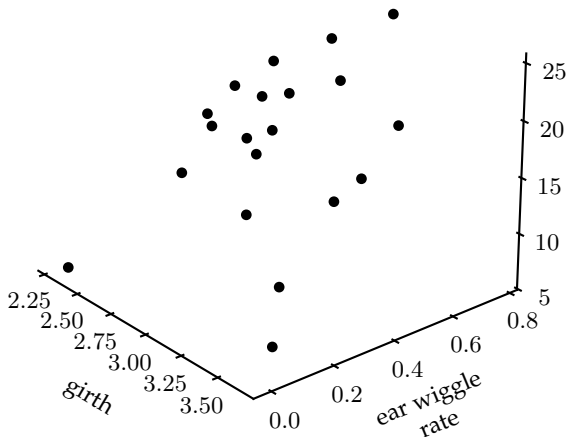
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New update term points toward origin

For this reason, ℓ_2 regularization often called weight decay

Ridge Regression Example

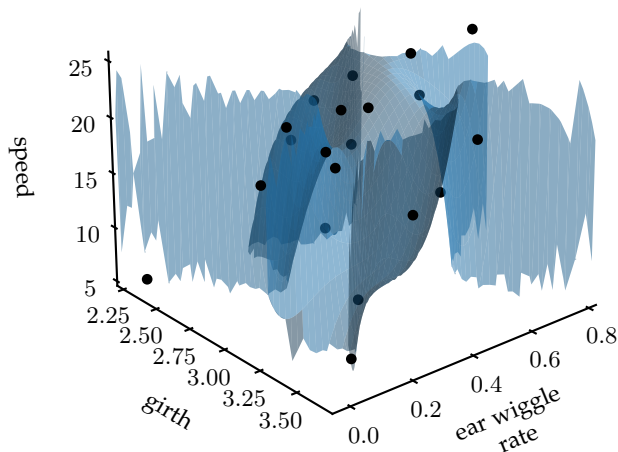
Data



Ridge Regression Example

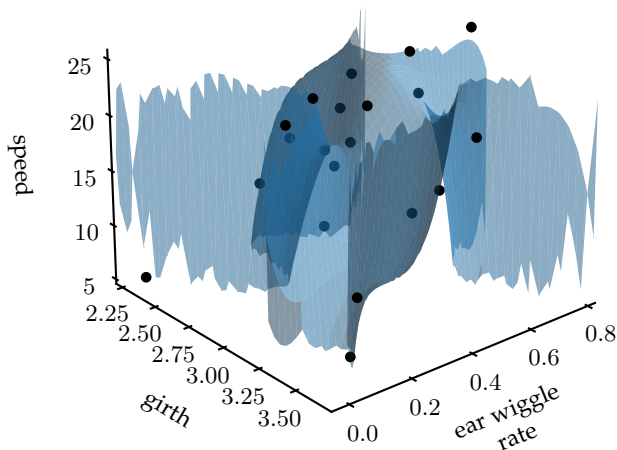
5th order polynomial ($n = 21$)

$\lambda = 0$



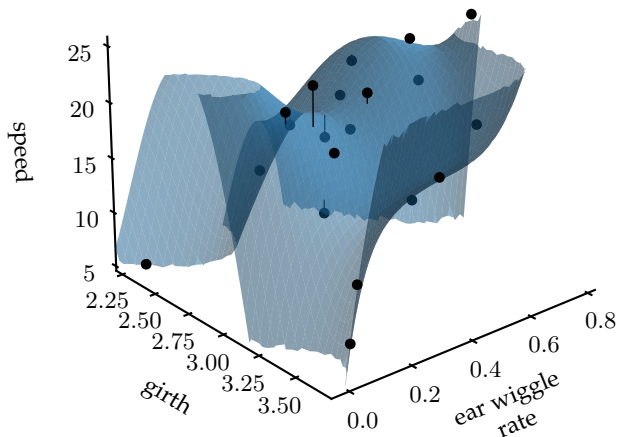
Ridge Regression Example

5th order polynomial ($n = 21$)
 $\lambda = 10^{-12}$



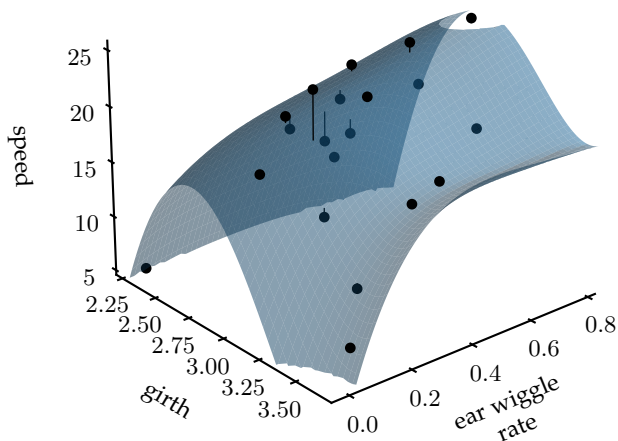
Ridge Regression Example

5th order polynomial ($n = 21$)
 $\lambda = 10^{-8}$



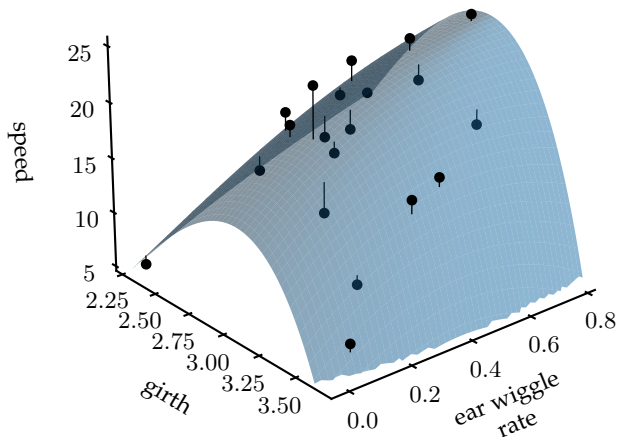
Ridge Regression Example

5th order polynomial ($n = 21$)
 $\lambda = 10^{-4}$



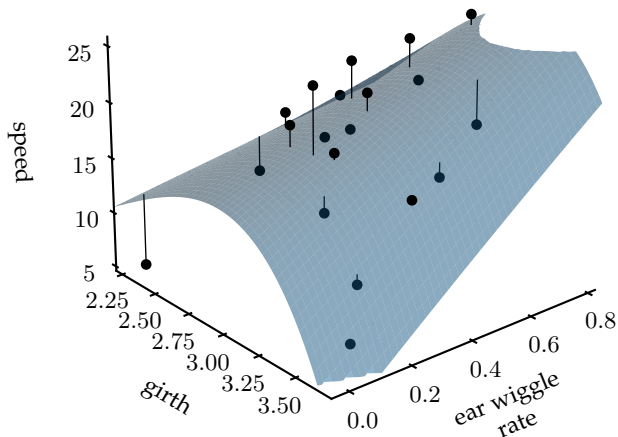
Ridge Regression Example

5th order polynomial ($n = 21$)
 $\lambda = 10^0$



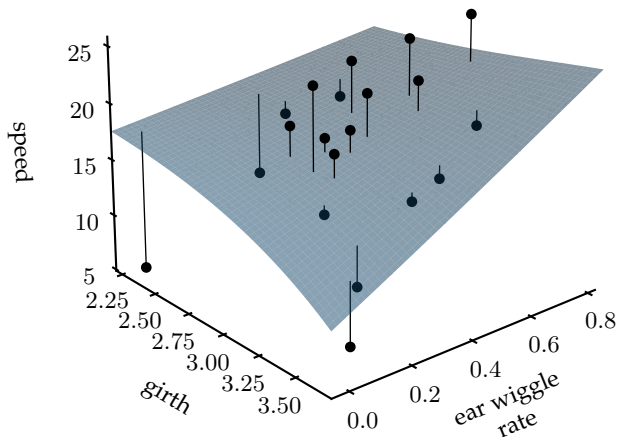
Ridge Regression Example

5th order polynomial ($n = 21$)
 $\lambda = 10^2$

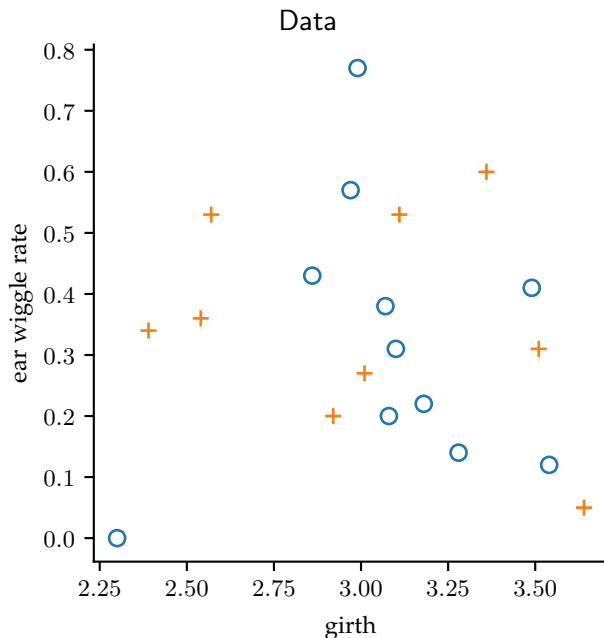


Ridge Regression Example

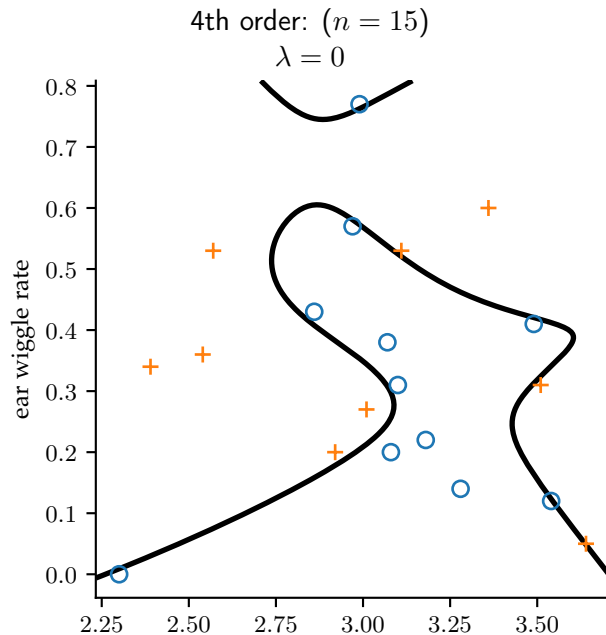
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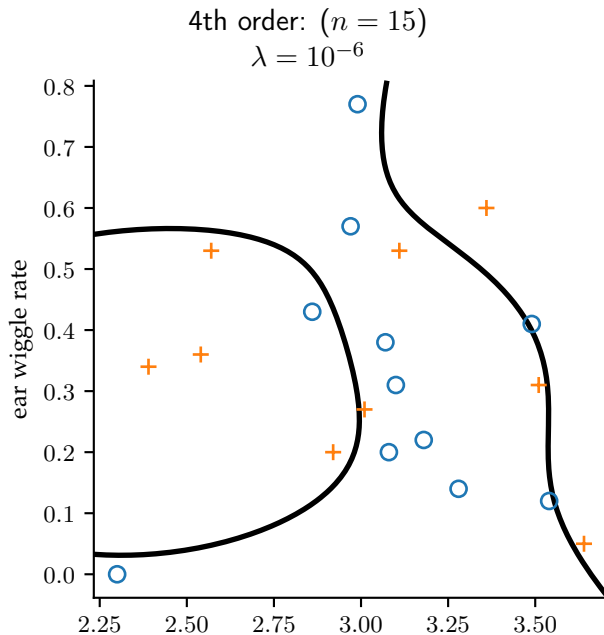
Logistic Regression with ℓ_2 Regularization Example



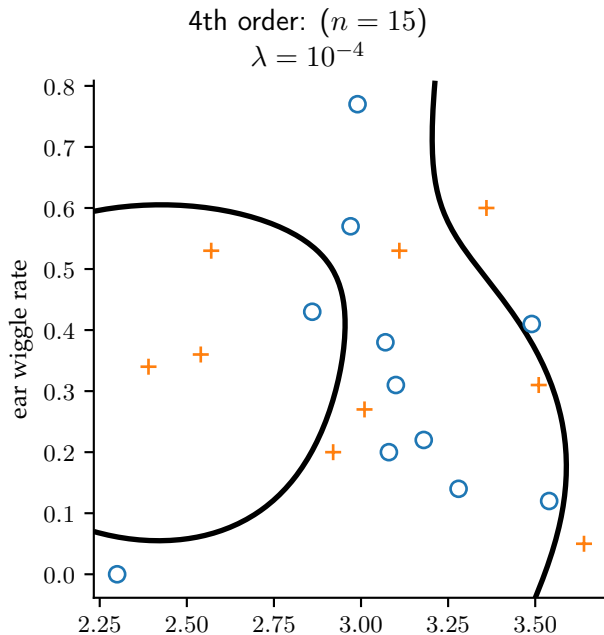
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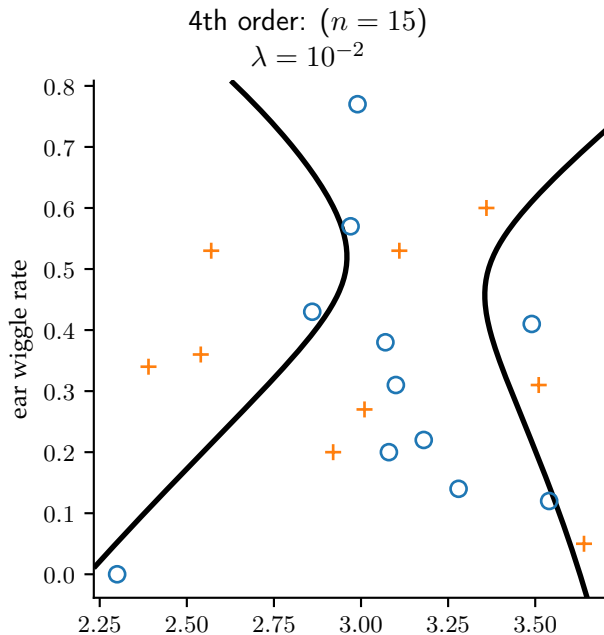
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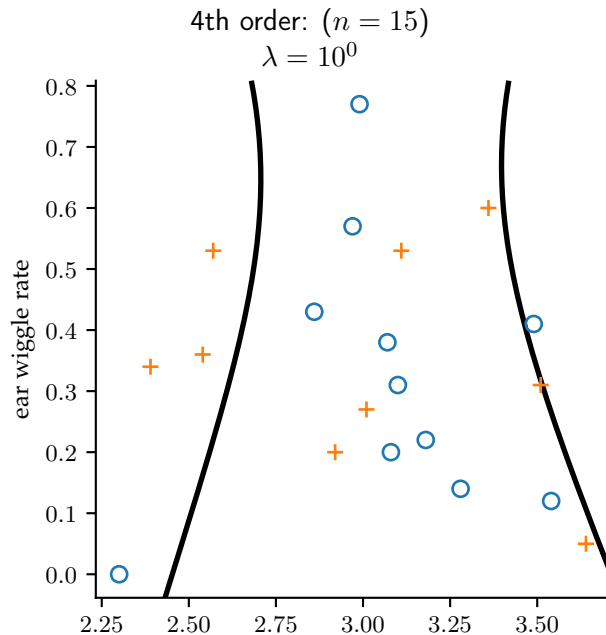
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Logistic Regression with ℓ_2 Regularization Example



Ridge Regression Effect

2 raw features, degree = 7, $m = 100$

